

Concurrency and Distributed Systems

Week 8 – Part 1: Debugging Evidence and Diagnosis

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Part 1 Roadmap

- 1 Midterm Frame
- 2 Thread Dump Evidence
- 3 Diagnosis Report

Week 8 Is a Checkpoint

Rubric

Real bug, mechanism, cost.

- A symptom is what the user sees.
- Evidence is what the system shows.
- Mechanism is why the failure happens.
- The fix must address the mechanism.

- Thread dump or simulated evidence.
- Diagnosis report.
- Fixed code.
- Performance tradeoff analysis.
- Five-minute technical explanation.

Evidence Excerpt

```
1 "request-worker-17" WAITING
2   at java.util.concurrent.CompletableFuture.join
3   at BrokenAsyncServer.handle
4
5 "dependency-client-3" TIMED_WAITING
6   at java.lang.Thread.sleep
7   at SlowDependency.call
```

Interpretation

The request path is waiting in a future join. The dependency path is delayed. Look for a missing timeout or missing completion path.

Reading Thread State

State	Debugging question
RUNNABLE	Is it CPU work or blocked native I/O?
BLOCKED	What monitor is contended?
WAITING	What completion or notification is missing?
TIMED_WAITING	Is it sleeping, timed wait, or timeout policy?

Report Structure

- ① Bug: the precise broken behavior.
- ② Evidence: the observation that proves the bug is plausible.
- ③ Mechanism: the causal path.
- ④ Fix: the smallest change that addresses the mechanism.
- ⑤ Residual risk: what remains uncertain.

Run

```
.\run-demo.ps1 01
```

```
.\run-demo.ps1 02
```

- Identify thread state.
- Identify top frame.
- Convert evidence into a diagnosis report.